

Question Paper

Exam Date & Time: 26-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIRST SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025
SUBJECT: CSE 1071/CSE_1071 PROBLEM SOLVING USING COMPUTERS

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) With a neat diagram of the computer organization, explain how each component coordinates together to make the computer system work. (3)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 1B) Analyze how expression evaluation changes if parentheses are added to the following line: (3)
 $\text{result} = 10 + 2 * 5 / 2 - 3;$
Discuss the above result before and after applying parentheses.
Explain the shortcut assignment operator in C expressions, with an appropriate example.

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 1C) Write a C program using a switch statement to check whether an entered alphabet is a vowel or a consonant. (4)
Analyze how using a switch simplifies the logic compared to multiple if-else conditions.

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 2A) Experiment with break and continue statements in terms of loop control and execution flow with an example. (2)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 2B) A library stores sorted book IDs. The librarian wants to search for a particular book ID quickly. (4)
Build a C program to search for a book ID using the appropriate searching technique. Also, justify why the corresponding searching technique is chosen to be appropriate over others.

Course Outcome: "CO3" / Cognitive Level: "Application"

- 2C) A teacher maintains the marks of students in two different tests as two matrices of size 3×3 . (4)
She wants to calculate the total marks for each student by adding the corresponding elements of the two matrices.
i) Develop a C program to add two 3×3 matrices and display the resulting matrix.
ii) Explain clearly how each element of the matrices is accessed and summed using nested loops.

Course Outcome: "CO3" / Cognitive Level: "Application"

- 3A) Develop a function named "difference" that takes two integers as input parameters and returns their absolute difference as an integer using C programming. Declare the function prototype and mention the formal parameters (2)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 3B) A positive integer is entered through the keyboard. Write a C function to obtain the prime factors of this number. For example, the prime factors of 24 are 2, 2, 2, and 3, and those of 35 are 5 and 7. (4)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 3C) Implement binary search recursively on a sorted array. Your recursive function should return the index of the key if found, otherwise -1. (4)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4A) A student writes a function **void increment (int a, int b)** to increase both variables by 10 and but finds the original values unchanged when he prints them in main. Analyze the error and write the c program by rewriting the function and print the updated value in main (Hint: use pointers). (3)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 4B) Design a C program to analyze student performance using nested function calls. (4)
Your program should:
i) Accept marks of N students in M subjects and store them in a 2D array.
ii) Compute each student's average using the function studentAverage().
iii) Call the function subjectAverage() inside studentAverage() to compute the average marks for each subject (demonstrating function nesting).
iv) Display both student-wise and subject-wise averages clearly.

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4C) Develop a C program that uses a structure to store the dimensions of a rectangle (length and width) and calculate and display its area and perimeter. (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

- 5A) Write a C program to define a structure to store census data (city name, population, literacy level), read data for 5 cities and display the city with the highest literacy level. (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

- 5B) Develop a C program to copy one string to another using pointers only. (without using library functions for strings) (4)

Course Outcome: "CO5" / Cognitive Level: "Application"

- 5C) Classify 4 cybercrimes where computers are used as a tool to commit crimes and explain 2 examples of such crimes. (3)

Course Outcome: "CO6" / Cognitive Level: "Understand"

-----End-----

Event Name	Subject
------------	---------

058-END SEMSTER-ODD-2025 CSE_1071:PROBLEM SOLVING USING COMPUTERS

#	Course Outcome	Description	Marks
1	1	CO 1	3
2	2	CO 2	9
3	3	CO 3	8

4	4	CO 4	17
5	5	CO 5	10
6	6	CO 6	3
Total			50

* NA - CO Not Selected

Event Name	Subject
------------	---------

058-END SEMSTER-ODD-2025CSE_1071:PROBLEM SOLVING USING
COMPUTERS

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	2	6(12%)
3	Application	10	34(68%)
4	Analysis	3	10(20%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)